



King Abdulaziz University

Faculty of Computing & Information Technology Computer Science Department

Exam 1 (Thursday February 27, 2020)
CPCS-223 (Analysis and Design of Algorithms)
Fall 2019
Points: 100 Time: 80 minutes

SOLUTION

Student ID: _____

Student Name: _____

*****Section:** _____ **WA**

Q	SO	Question	Max. Marks	Obt. Marks
01	A	1	15	
02	A	2	15	
03	A	3	20	
04	B	4	10	
05	B	5	20	
06	A	6	20	

Instructor:

☐ Dr. Mahar

☐ Dr Muhammad Umair

CLO – 1
Question No. 1

SO – A
(15) points

PART – A

What is an “Algorithm”? Write down the main characteristics of an Algorithm.

Solution:

Algorithm Definition:

(05 points)

An algorithm is a well-defined computational procedure that takes some values or set of values as input and produces some values or set of values as output.

Characteristics:

(10 points)

S. No.	Characteristics
01	It accepts one or more inputs
02	It returns at least one output
03	It terminates after finite steps
04	It is independent of a specific programming language, machine or compiler
	Unlike the computer programs (which are for computers understanding), algorithms are understood by humans
	for every input it halts with correct output

CLO – 2

SO – A

Question No. 2

15 points

On which parameters the efficiency of an algorithm depends? Describe the main characteristics of “Time Efficiency” for an algorithm.

Solution:

Efficiency Parameters:

(05 points)

1);

Space Efficiency

2);

Time Efficiency

Time Efficiency:

(10 points)

S. No.	Characteristics
1	Determines how fast an algorithm solves a given problem.
2	It is the measure of time taken by the algorithm to produce the output with a given input
3	The time complexity can depend on more than one parameter. For example, the running time of a graph algorithm can depend on both the number of vertices and edges in the graph
4	Assuming that an algorithm takes $T(n)$ time to process n data items. The function $T(n)$ is referred to as running time of the algorithm
	It would be seen that the time efficiency is the most significant metric for algorithm analysis. For this reason the main focus of an algorithm analysis would be on determining time complexity

CLO – 3
Question No. 3

SO – A
(05+ 15) points

PART – A

Prove that

$$2n^2 + 3n + 6 = \Theta(n^3)$$

Solution:

$2n^2 + 3n + 6 = \Theta(n^3)$	
$c_2(n^3) \leq 2n^2 + 3n + 6 \leq c_1(n^3)$	
$c_2(n^3) \leq 2n^2 + 3n + 6$	$2n^2 + 3n + 6 \leq c_1(n^3)$
$c_2 n \leq 2 + 3/n + 6/n^2$	$2 + 3/n + 6/n^2 \leq c_1(n^3)$
Minimum value = 2 It is not possible for any value of c_2 that the above relation holds. So it is disproved	Max value = 11 For $c_1 \geq 11$ and $n_0 = 1$ It is proved

Hence

$$2n^2 + 3n + 6 \neq \Theta(n^3)$$

PART – B

Find the recurrence of the following given algorithm to determine the number of moves for “n” disks and solve it using **any method** you have studied in your class for the addition operations.

Tower(N, BEG, AUX, END)

1. If N= 1 Then
 1. Move Disk from “Beg” to “End”
 2. Return
2. Tower (N-1, Beg, End, Aux)
3. Move Disk from “Beg” to “End”
4. Tower (N-1, Aux, Beg, End)
5. Return

Solution:

Recurrence:

(03 points)

$$\mathbf{M(n) = 2 M(n-1) + 1}$$

Recurrence Solution:

(12 points)

$$M(n) = 2M(n-1) + 1 \quad \text{sub. } M(n-1) = 2M(n-2) + 1 \quad |$$

$$= 2[2M(n-2) + 1] + 1 = 2^2M(n-2) + 2 + 1$$

$$\text{sub. } M(n-2) = 2M(n-3) + 1$$

$$= 2^2[2M(n-3) + 1] + 2 + 1 = 2^3M(n-3) + 2^2 + 2 + 1.$$

the next one will be |

$$2^4M(n-4) + 2^3 + 2^2 + 2 + 1$$

after i substitutions,

$$M(n) = 2^i M(n-i) + 2^{i-1} + 2^{i-2} + \cdots + 2 + 1$$

$$= 2^i M(n-i) + 2^i - 1.$$

after i substitutions,

$$\begin{aligned} M(n) &= 2^i M(n-i) + 2^{i-1} + 2^{i-2} + \cdots + 2 + 1 \\ &= 2^i M(n-i) + 2^i - 1. \end{aligned}$$

For real $x \neq 1$, the summation

$$\sum_{k=0}^n x^k = 1 + x + x^2 + \cdots + x^n$$

Since the initial condition is specified for $n = 1$, which is achieved for $i = n - 1$,

$$\begin{aligned} M(n) &= 2^{n-1} M(n - (n - 1)) + 2^{n-1} - 1 \\ &= 2^{n-1} M(1) + 2^{n-1} - 1 \\ &= 2^{n-1} + 2^{n-1} - 1 \\ &= 2^n - 1. \end{aligned}$$

$$\sum_{k=0}^n x^k = \frac{x^{n+1} - 1}{x - 1}$$

$$M(n) = \Theta(2^n)$$

CLO – 4

SO – B

Question No. 4

10 points

Find the running time of the following algorithm using **any technique** which you studied in your class.

ALGORITHM *MatrixMultiplication*($A[0..n-1, 0..n-1]$, $B[0..n-1, 0..n-1]$)

//Multiplies two n -by- n matrices by the definition-based algorithm

//Input: Two n -by- n matrices A and B

//Output: Matrix $C = AB$

for $i \leftarrow 0$ **to** $n-1$ **do**

for $j \leftarrow 0$ **to** $n-1$ **do**

$C[i, j] \leftarrow 0.0$

for $k \leftarrow 0$ **to** $n-1$ **do**

$C[i, j] \leftarrow C[i, j] + A[i, k] * B[k, j]$

return C

Solution:

Using Ram Model

$$\begin{array}{ll}
 \text{for } i \leftarrow 0 \text{ to } n-1 \text{ do} & n+1 \\
 & n-1 \\
 \text{for } j \leftarrow 0 \text{ to } n-1 \text{ do} & \sum_{i=0}^{n-1} (n+1) \\
 & n-1 \quad n-1 \\
 C[i, j] \leftarrow 0.0 & \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} 1 \\
 & n-1 \quad n-1 \\
 \text{for } k \leftarrow 0 \text{ to } n-1 \text{ do} & \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} (n+1) \\
 & n-1 \quad n-1 \quad n-1 \\
 C[i, j] \leftarrow C[i, j] + A[i, k] * B[k, j] & \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} \sum_{k=0}^{n-1} 1
 \end{array}$$

$$T(n) = (n+1) + \sum_{i=0}^{n-1} (n+1) + \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} 1 + \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} (n+1) + \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} \sum_{k=0}^{n-1} 1$$

$$T(n) = (n+1) + n(n+1) + n * n + n * n(n+1) + n * n * n$$

$$T(n) = n+1 + n^2 + n + n^2 + n^3 + n^2 + n^3$$

$$T(n) = 2n^3 + 3n^2 + 2n + 1$$

$$T(n) = \Theta(n^3)$$

Informal Analysis

Outermost loop executes: n times

Middle loop executes: n times

Inner loop executes: n times

$$T(n) = n * n * n = O(n^3)$$

CLO – 5**Question No. 5**

For the given data set, how the arrays A will be updated during execution of the given algorithm of “Question05”? You are required to reflect all changes to these arrays for each iteration.

SO – B**20 points****ALGORITHM 05 (A [0.. n-1])**

```

int MinPos, temp, i, j;
for(i = n-1; i > 0; i--)
    MinPos = i;
    for ( j = 0; j < i; j++ )
        If ( A [ j ] < A [ MinPos ] ) MinPos = j;
    Temp = A [ i ];
    A [ i ] = A [ MinPos ];
    A [ MinPos ] = temp;

```

Solution:**A**

0	1	2	3	4	5	6	7	8	9
49	68	5	61	35	57	6	44	99	91

Iteration	0	1	2	3	4	5	6	7	8	9
1	49	68	91	61	35	57	6	44	99	5
2	49	68	91	61	35	57	99	44	6	5
3	49	68	91	61	44	57	99	35	6	5
4	49	68	91	61	99	57	44	35	6	5
5	57	68	91	61	99	49	44	35	6	5
6	99	68	91	61	57	49	44	35	6	5
7	99	68	91	61	57	49	44	35	6	5
8	99	91	68	61	57	49	44	35	6	5
9	99	91	68	61	57	49	44	35	6	5
10 (Final Values)	99	91	68	61	57	49	44	35	6	5

CLO – 6
Question No. 6

SO – A
(20) points

Write the pseudo-code / algorithm to calculate the sum of all “**even digits**” in a given number.

Input: a number from user

e.g. 348765

Process: separate the digits of the number and add them up

Output: return the sum value

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(i.e. 4+8+6)

Solution:

```
public static int sumDigits(int n) {  
  
    if ( n == 0 )  
  
        return 0;  
  
    else if ((n%10) % 2 == 0)  
  
        return sumDigits(n/10) + (n%10);  
  
    else  
  
        return sumDigits(n/10)  
}
```

Extra Sheet for rough work